

Vulnerability Assessment Report

1st January 20XX

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 20XX to August 20XX. [NIST SP 800-30 Rev. 1](#) is used to guide the risk analysis of the information system.

Purpose

The database server is valuable to the business because it stores and manages important company data that is needed for daily operations. Securing the data is important because unauthorized access, modification, or loss of information could affect business operations, customer trust, and the confidentiality and integrity of company information. If the server were disabled, employees and other systems might lose access to critical data and applications, which could interrupt normal business activities and potentially result in financial loss.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
<i>hacker</i>	<i>Obtain sensitive information via exfiltration</i>	3	3	9
<i>employee</i>	<i>Change or delete information</i>	2	3	6
<i>System administrator</i>	<i>Disrupt mission-critical operations</i>	2	3	6

Approach

The three threat sources and events were selected because the database server stores important business information and is connected to other systems on the network. A hacker could steal sensitive information through exfiltration, while an employee could alter or delete critical information using legitimate access. A system administrator could also disrupt mission-critical operations through misuse or accidental changes because of their privileged access. These threats are significant business risks because they could cause data loss, operational disruptions, financial damage, and loss of customer trust.

Remediation Strategy

The organization should implement **multi-factor authentication (MFA)** and the **principle of least privilege** to prevent unauthorized access to the database and reduce the risk of sensitive information being stolen or altered. The **AAA framework** should be used to strengthen authentication, authorization, and accounting by monitoring user activity and restricting access to critical data. **Defense in depth**, including firewalls, network segmentation, encryption, and regular security updates, can reduce the impact of attacks and help protect mission-critical operations. Regular access reviews and vulnerability scans should also be performed to identify and address new risks.